A5

Equal-area cells for geographic data.

Install & load

```
INSTALL a5 FROM community;
LOAD a5;
```

01 At a glance

Turn longitude and latitude into pentagonal cell IDs for grouping, joining, and geographic rollups. A5 uses equal-area cells at each resolution, with integer IDs and functions for hierarchy, neighborhoods, and geometry conversion.

Longitude / latitude → A5 cell ID → Group / join / roll up

02 Core SQL surface

`a5_lonlat_to_cell(lon,lat,res)`

Map longitude, latitude in degrees to a UBIGINT cell ID.

`a5_cell_to_lonlat(cell)`

Return the cell center as [longitude, latitude], not the original point.

`a5_cell_to_parent(cell,res)`

Find a coarser ancestor for hierarchical rollups.

`a5_cell_area(res)`

Return equal cell area in square meters at a resolution.

`a5_grid_disk(cell,k)`

Return cells within k edge steps, including the center cell.

`a5_cell_to_geometry(cell)`

Produce a native GEOMETRY polygon for mapping.

`a5_geometry_to_cells(geom,res)`

Cover geometry; polygons return a compacted covering.

03 Keep in mind

Coordinates and identifiers

Pass longitude first, latitude second, in degrees. Resolutions run from 0 to 30; larger values mean smaller cells. Preserve IDs as UBIGINT or use a5_u64_to_hex when exporting to systems with limited integer precision.

04 Encode New York City

```
SELECT a5_lonlat_to_cell(
  -74.0060, 40.7128, 15
) AS cell;
SELECT a5_cell_area(15) AS area_m2;
```

Cell 2760559170984869888; area ≈31,669 m².

05 Count observations by cell

```
WITH points(lon,lat) AS (VALUES
  (-74.0060,40.7128), (-74.0061,40.7129),
  (-73.9857,40.7484)
)
SELECT a5_lonlat_to_cell(lon,lat,15) AS cell,
       count(*) AS n
FROM points
GROUP BY cell;
```

Equal-resolution cell IDs are ordinary grouping and join keys.

06 Roll up and draw a cell

```
WITH p AS (
  SELECT a5_lonlat_to_cell(
    -74.0060,40.7128,15) AS cell
)
SELECT a5_cell_to_parent(cell,10) AS parent,
       a5_cell_to_geometry(cell) AS polygon
FROM p;
```

The parent is a hierarchy key; the polygon is native GEOMETRY.

07 In practice

Render at one resolution

Coverings from a5_geometry_to_cells, a5_grid_disk, and a5_compact end in a compaction marker that is not a cell. Expand with a5_uncompact(cells) before unnesting or drawing them. A5 1.0 changed cell IDs: convert stored pre-1.0 IDs once with a5_migrate.

Full docs & examples ↗ Source ↗